1

General Session

Prescription predicament: Unveiling the financial fallout of drug shortages.

Puneeth Indurlal, Jody S. Garey, John Albaugh, Arran Standring, Jessica White, Leslie T. Busby, Lalan S. Wilfong; The US Oncology Network, The Woodlands, TX; McKesson, Irving, TX; RMCC/US Oncology, Boulder, CO; Texas Oncology, The US Oncology Network, Dallas, TX

Background: Managing drug shortages has become a part of care delivery in oncology. Shortages have been reported to result in missed/delayed treatment, substitution of alternative therapy, increased costs, and significant administrative burden. A national shortage of cisplatin caused by manufacturing disruptions was reported by the Food and Drug Administration in the United States from February until August 2023 (shortage period). We assessed the effects of cisplatin shortage on the treatment of head and neck cancer (HNC) at practices in The US Oncology Network (The Network). **Methods:** Using drug administration data from medical records and claims for 26 practices in The Network prior to (July 2022 to January 2023), during, and after (September 2023 to March 2024) the shortage period, we assessed the utilization trends and financial impact (using Medicare average sale price (ASP)) of cisplatin and other anti-cancer drugs for HNC patients. **Results:** During the shortage period, cisplatin utilization for HNC decreased by 15% compared to pre-shortage levels. The lowest utilization occurred in June and July 2023, with a 60% decrease. Practices initially obtained supplies through an allocation model and fair-share utilization guidance from The Network Pharmacy & Therapeutics Committee. During the shortage period, alternative chemotherapy drugs for HNC saw increased use: carboplatin by 40%, paclitaxel by 24%, 5-fluorouracil by 5.3%, and cetuximab by 15%. The increased use of alternative drugs corresponded to the decrease in cisplatin usage, with new starts shifting to carboplatin (with or without paclitaxel), cetuximab and 5-fluorouracil. Ten percent of existing cisplatin recipients were shifted to an alternative drug during the shortage period. Interestingly, 5-fluorouracil was also in short supply, albeit less acutely than cisplatin. After the shortage, cisplatin volumes rebounded by 8% of pre-shortage use, while carboplatin use dropped below pre-shortage levels. Cetuximab use remained consistently 12% higher. The average cost per administration was $18 for cisplatin, $14 for carboplatin, $16 for paclitaxel, $22 for 5-fluorouracil, and $2,607 for cetuximab, based on the ASP. The increased use of cetuximab instead of cisplatin resulted in a 16% total cost increase, leading to a 144-fold increase in costs at the administration level, impacting payer costs and patient cost-sharing amounts. **Conclusions:** Cisplatin shortage led to a shift in utilization to alternative therapies for HNC, resulting in significant cost increases for both payers and patients. These consequences may not be limited to specific cancer types or drugs and may have other undesirable effects. Even temporary shortages of drugs can lead to lasting utilization changes and have long-term financial implications. The impacts of drug shortages on treatment decisions, guideline concordance, costs, and patient outcomes are apt for further investigation. Research Sponsor: None.